Experiment 5:

Calculation and Verification of Frequency Deviation and Modulation Index of FM Signals Using MATLAB Simulation

Aim: To calculate the frequency deviation and modulation index of Frequency Modulated (FM) signals using MATLAB simulation.

Software Required:

MATLAB

Program:

```
clc; clear; close all;

Fs=100000; T=0.02;
t=0:1/Fs:T-1/Fs;
Ac=5; Am=2; fc=10000; fm=500; kf=500;

df=kf*Am;
beta=df/fm;

msg=Am*cos(2*pi*fm*t);
c=Ac*cos(2*pi*fc*t);
FM=Ac*cos(2*pi*fc*t+beta*sin(2*pi*fm*t));
fth=fc+df*cos(2*pi*fm*t);

z=hilbert(FM);
fest=[diff(unwrap(angle(z)))*Fs/(2*pi),NaN];
v=101:length(t)-100;
fmin=min(fest(v)); fmax=max(fest(v));

fprintf('Frequency deviation = %.2f Hz\n',df);
fprintf('Modulation index = %.2f\n',beta);
fprintf('Minimum frequency = %.2f Hz\n',fc-df);
fprintf('Maximum frequency = %.2f Hz\n',fc+df);
fprintf('Estimated deviation = %.2f Hz\n',(fmax-fmin)/2);

n=round(0.006*Fs);
figure;
subplot(3,1,1); plot(t(1:n),msg(1:n),'b'); grid on;
title('Message Signal'); ylabel('Amplitude (V)');

subplot(3,1,2); plot(t(1:n),c(1:n),'k'); grid on;
title('Unmodulated Carrier Signal'); ylabel('Amplitude (V)');

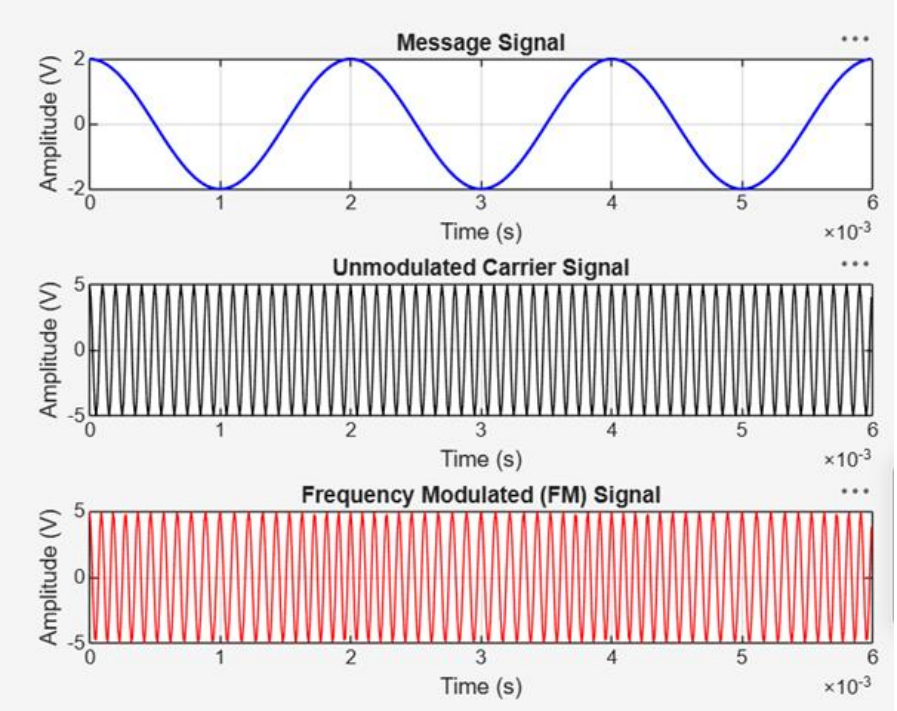
subplot(3,1,3); plot(t(1:n),FM(1:n),'r'); grid on;
title('Frequency Modulated (FM) Signal');
xlabel('Time (s)'); ylabel('Amplitude (V)');
```

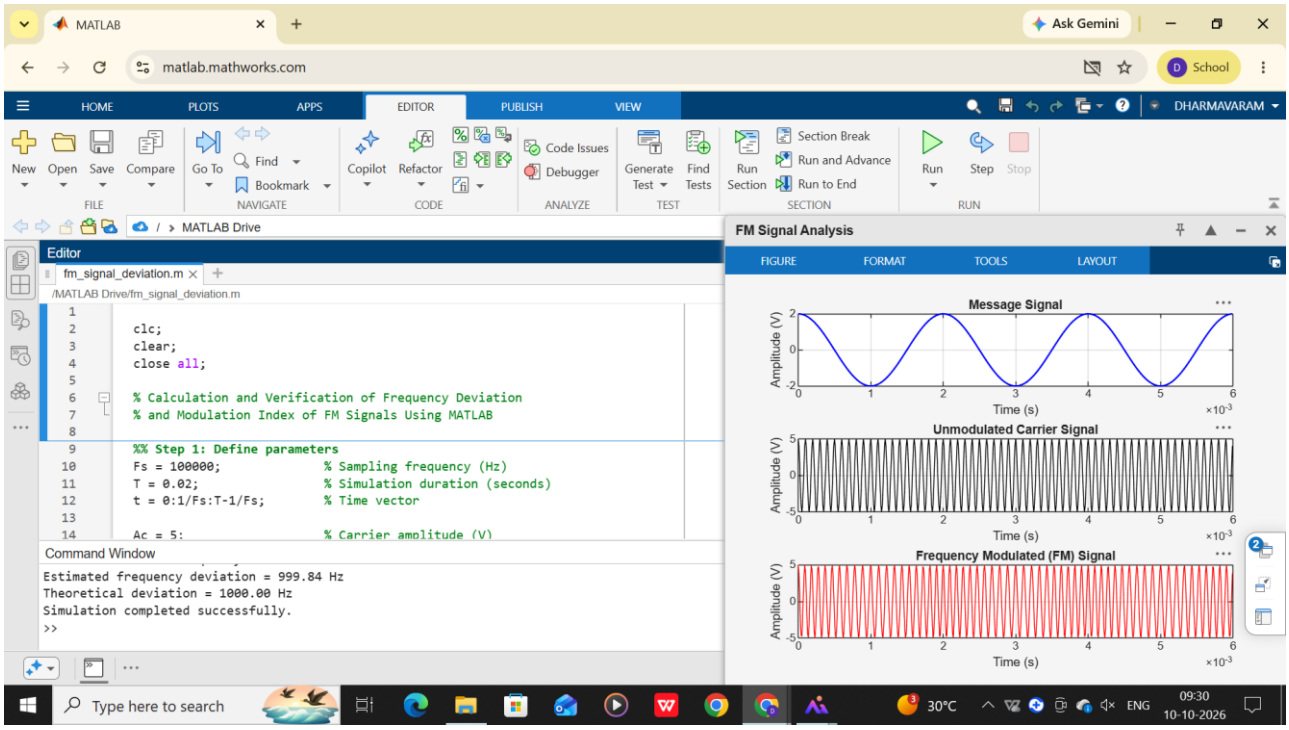
```
figure;
plot(t,fth,'b',t,fest,'r--',t,fc+zeros(size(t)),'k:');
grid on;
title('Instantaneous Frequency Verification');
xlabel('Time (s)'); ylabel('Frequency (Hz)');
legend('Theoretical','Estimated','Carrier');

N=length(FM);
Y=abs(fft(FM))/N;
P=Y(1:floor(N/2)+1);
P(2:end-1)=2*P(2:end-1);
f=(0:floor(N/2))*Fs/N;

figure;
plot(f/1000,P,'m'); grid on;
title('FM Frequency Spectrum');
xlabel('Frequency (kHz)'); ylabel('Amplitude');
xlim([5 15]);
```

Output:





Result: The FM signal was successfully generated and analyzed using MATLAB. The frequency deviation and modulation index were calculated and verified through simulation.